Year 10 Science
Physical Sciences
Worksheet - Speed and Velocity

SPEED

The speed of a body is the rate of change of distance by the body.

The average speed of a body over a distance travelled in a time interval can be determined by:

$$\text{Speed (av)} = \frac{d}{t}$$

where d = distance (measured from zero)
t = time interval (measured from zero)

Speed is a scalar quantity as it involves no specific direction. Distance is the total length of the path covered by the body.

If to accurately describe motion direction is required, then *VELOCITY* is used in place of speed.

VELOCITY

The velocity of a body is the rate of change of displacement of the body.

The average velocity of a body over a displacement in a measured time interval can be determined by:

$$v_{av} = \frac{s}{t}$$

v_{av} = average velocity
s = displacement (measured from zero)
t = time interval (measured from zero)

Velocity is a vector quantity. *The direction of the velocity is that of the displacement (unless otherwise stated).*

UNITS OF VELOCITY AND SPEED

Velocity and speed are both determined by dividing units of length (m) by units of time (s).

Units: $\frac{\text{metres}}{\text{seconds}} = \frac{m}{s} = m\ s^{-1}$ (or m/s)

DETERMINATION OF DISPLACEMENT

The equation for average velocity can be rearranged to determine the displacement.

$$v_{av} = \frac{s}{t}$$

rearranging gives: $s = v_{av} \times t$

similarly $d = \text{speed (av)} \times t$

TYPE EXAMPLE 7: Calculate the average velocity of a vehicle which travels 3 km east in 5 minutes.

$v_{av} = ?$
$s = 3\ km = (3 \times 1\,000)\ m$
$t = 5\ minutes = (5 \times 60)\ s$

$$v_{av} = \frac{s}{t} = \frac{3 \times 1\,000}{5 \times 60} = 10\ m\ s^{-1}$$

i.e. the average velocity is 10 m s^{-1} east.

TYPE EXAMPLE 8: Determine the displacement of a body which moves with a uniform velocity of 2.4 m s^{-1} for 50 s.

$v_{av} = 2.4 \text{ m s}^{-1}$
$s = ?$
$t = 50 \text{ s}$

$$v_{av} = \frac{s}{t}$$
$$\therefore \quad s = v_{av} \times t$$
$$= 2.4 \times 50$$
$$= 120 \text{ m}$$

i.e. the displacement is 120 m in the direction of the velocity.

SET 6: VELOCITY I

1. What is the average speed of a sprinter who runs 100 m in 12.5 s?
2. A car takes 80 s to complete one lap of a 2 km circuit. What is its average speed (in m s^{-1})?
3. A body travels 36 m in 4 s. What is its average speed?
4. How far will a bird fly in 25 s travelling with an average speed of 6 m s^{-1}?
5. A person swims from one end of a pool to the other at an average velocity of 1.5 m s^{-1} in 20 s. What is the length of the pool?
6. Determine the distance travelled by a plane in 1 hour if it flies due south with a uniform velocity of 90 m s^{-1}
7. How long will it take for a body moving with a uniform velocity of 7 m s^{-1} to cover 98 m?
8. A cricket ball is thrown a distance of 90 m from the boundary to the wicket at an average horizontal speed of 30 m s^{-1}. How long will it take for the ball to reach the wicket?
9. Determine the time taken for a horse running with an average speed of 14 m s^{-1} to complete one circuit of a 840 m track.
10. How long will it take to complete a trip from Perth to Kalgoorlie, a distance of 595 km, travelling at an average speed of 70 km h^{-1}?
11. A train moving with an average speed of 80 km h^{-1} takes 0.25 hours to travel from one railway siding to the next. What is the distance between the two sidings?
12. Determine the average speed (km h^{-1}) of a plane which flies a distance of 720 km in 1.5 hours?

SET 6

1. 8 m s^{-1} **2.** 25 m s^{-1} **3.** 9 m s^{-1} **4.** 150 m **5.** 30 m
6. 324 000 m **7.** 14 s **8.** 3 s **9.** 60 s **10.** 8.5 h
11. 20 km **12.** 480 km h^{-1}

SET 7: VELOCITY II

1. An athlete, commencing at a point due east, completes half a lap of a 440 m track in 25 s. Determine the athlete's:
   (a) average speed.
   (b) average velocity (in westerly direction).
   (c) average velocity if the athlete completes one full lap at the same pace.
2. A car travels with an average speed of 20 m s^{-1} due north for 3 km then due west for another 4 km calculate:
   (a) the total time taken for the journey.
   (b) the displacement of the car.
   (c) the average velocity of the car.
3. A person walks for 10 s at 1 m s^{-1} then for another 10 s in the same direction at 2 m s^{-1}. Find the:
   (a) total displacement.
   (b) average velocity.
4. A body travels 200 m at a uniform velocity of 4 m s^{-1} and a further 150 m in the same direction at a uniform velocity of 7.5 m s^{-1}. Determine the:
   (a) total time taken.
   (b) average velocity for the whole journey.
5. A cyclist travels at 6 m s^{-1} S for 10 minutes and then returns 600 m along the same path. What is the final displacement of the cyclist?
6. In a 50 m swimming race the winner completed the race in 32 seconds. By what margin (length) did the winner defeat the second placegetter who swam with an average velocity of 1.5 m s^{-1}?
7. A cyclist moving uniformly at 5 m s^{-1} overtakes a pedestrian walking in the same direction in 6 s. If the pedestrian initially was 18 m ahead of the cyclist, find the pedestrian's average velocity.
8. Determine the time saved by travelling 8 km at 40 m s^{-1} rather than at 32 m s^{-1}.
9. How far will a car moving at 20 m s^{-1} travel in the same time that a car moving at 25 m s^{-1} travels 1 km?
10. A car travelling at 15 m s^{-1} is 10 m behind another vehicle moving in the same direction. If it takes 10 s to overtake this vehicle, what is the speed of the slower vehicle?

SET 7

1. (a) 8.8 m s^{-1} (b) 5.6 m s^{-1} W (c) 0 **2.** (a) 350 s (b) 5 km N 53° W
(c) 14.3 m s^{-1} N 53° W **3.** (a) 30 m (b) 1.5 m s^{-1} **4.** (a) 70 s (b) 5 m s^{-1}
5. 3 000 m S **6.** 2 m **7.** 2 m s^{-1} **8.** 50 s **9.** 800 m **10.** 14 m s^{-1}